ICONHIC2019

A hybrid model for the flood prediction in Mohawk River, New York

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Geology Environment Sustainability

INTRODUCTION - METHODS

- Flood events are dependent on climatic and hydrogeological variables
- Time series data consists of different signals which they need to be separated to reveal the “real” peaks in flooding.

METHODS

Time series decomposition

$$X(t) = L(t) + Se(t) + Sh(t)$$

X(t) represents the original time series of a variable,

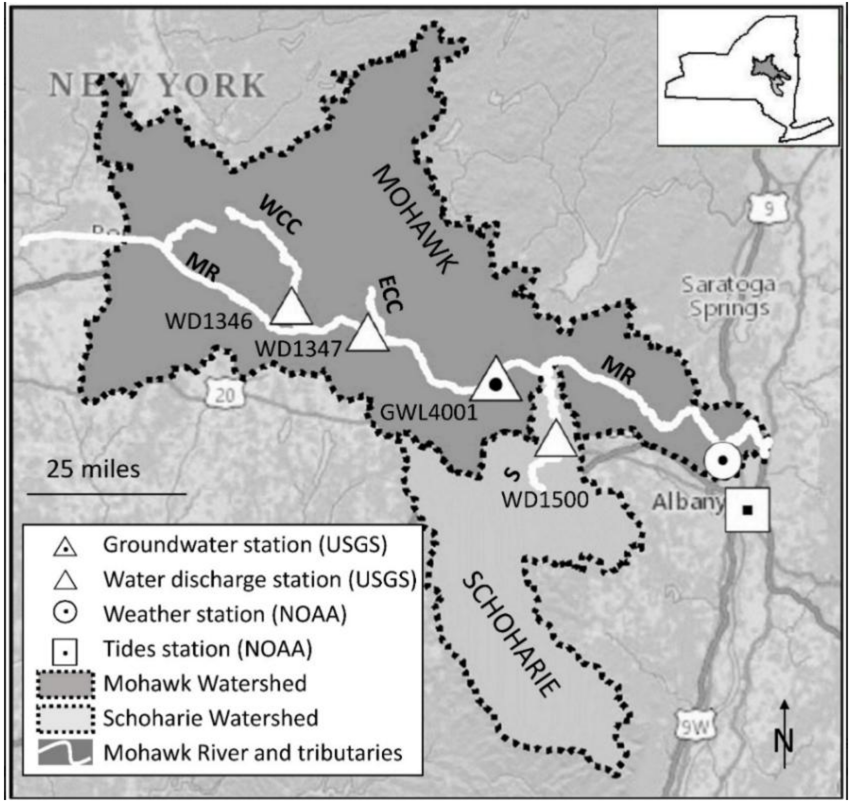
L(t) is the long-term trend component,

Se(t) is the seasonal component, and Sh(t) is the short-term component.

METHODS - DATA

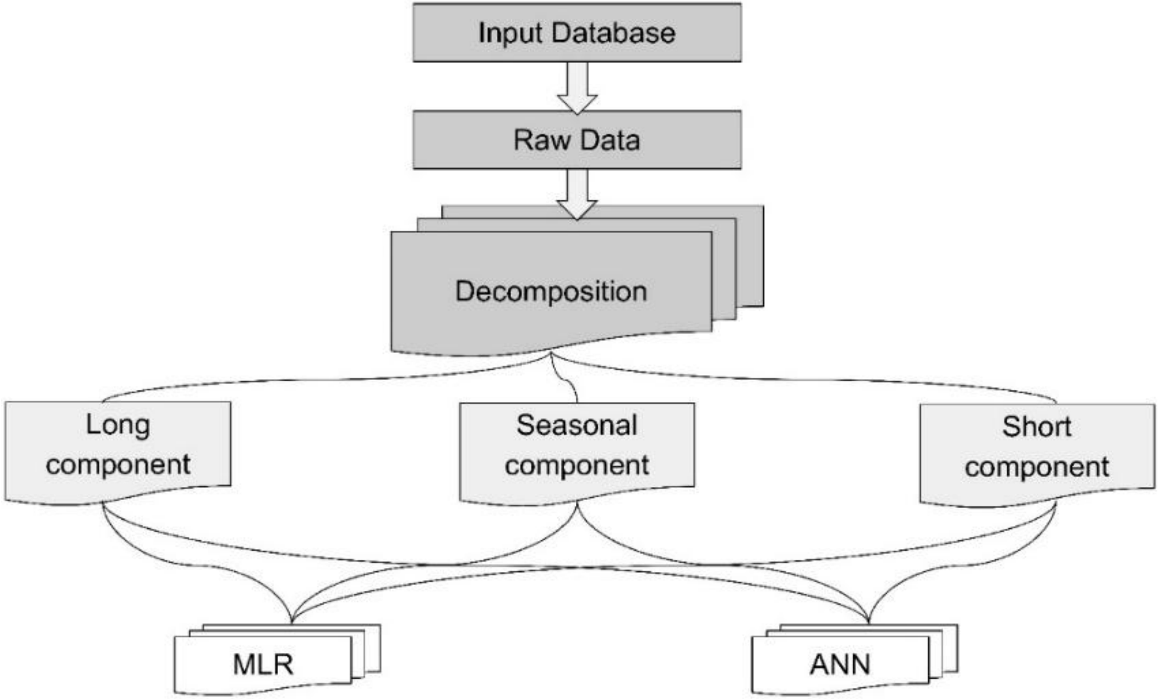
Climatic and hydrogeological data originated from:

- **United States Geological Survey** (USGS) and
- **National Oceanic and Atmospheric Administration** (NOAA)



METHODS

The hybrid model that consists of the data processing and the integration of the decomposition into the multiple linear regression (MLR) or artificial neural network (ANN) application.



RESULTS

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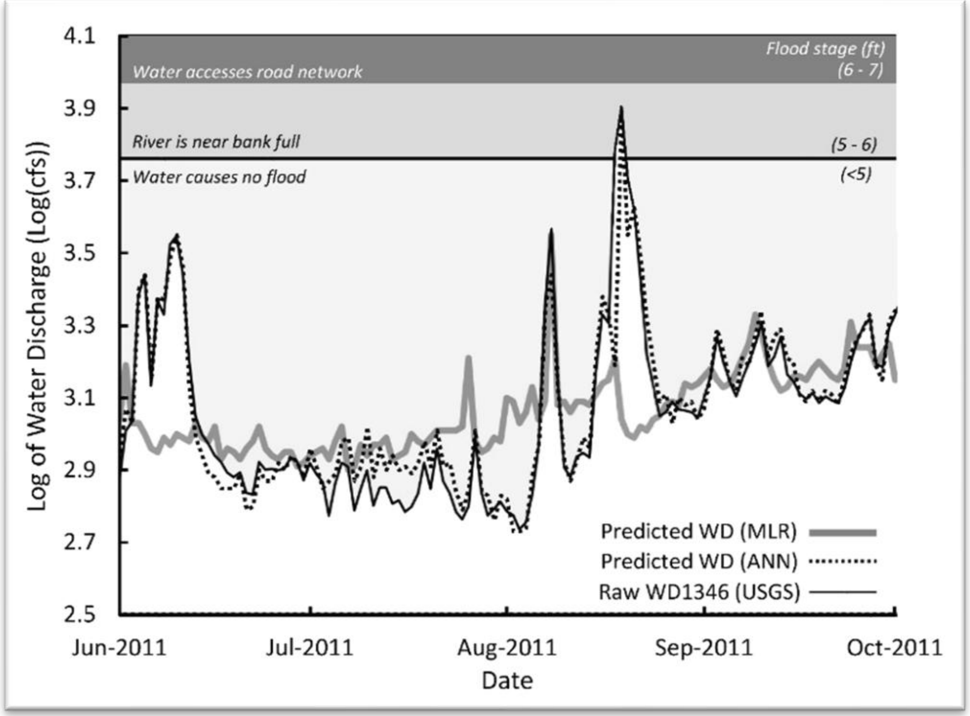
- ANN prediction up to 95.8 %
- MLR prediction up to 79.6 %

Station	Long	Seasonal	Short	Total Explanation
Variance %				
WD1346	23.5	37.1	39.4	
WD1347	23.4	36.1	40.5	
WD1500	53.1	16.2	30.7	
MLR Explanation %				
WD1346	17.3	34.5	27.8	79.6
WD1347	17.2	33.2	27.7	78.1
WD1500	44.1	14.8	10.5	69.4
ANN Explanation %				
WD1346	23.0	35.4	37.4	95.8
WD1347	22.8	34.5	38.6	95.8
WD1500	51.0	15.2	28.9	95.1



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CONCLUSIONS

- The overall forecasting accuracy has been increased by up to six times using the Artificial Neural Networks compared with the Multiple Linear Regression models.
- For station WD1500, the overall explanation has been increased six times (from 69.4% to 95.1%) using ANNs. Similar results can be conducted for the remaining stations.
- The short-term component describes short-term variations which can be modeled with a better accuracy using a probabilistic method (ANN).
- The explanation of all the stations varies from 95.1% to 95.8% which allows a very accurate prediction of the flood events.